DESIGN LAB

Phase 7 Final Technical Report

Inferential Validation of the Candidate Gate 5 Architecture

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Prince Upadhyay
Independent Research
starxshark@gmail.com
GitHub: reggaesharkk

FINAL STATUS

All 10 production blocks recovered and verified. 4,000 GLMM fits. Final aggregation integrity pass. Under the locked Phase 7 decision rule, M1 is the candidate architecture; M2 is not promoted.

Executive summary

Phase 7 completed the prespecified inferential-validation experiment after recovery from a documented R-environment dependency failure. The final production set contains 4,000 GLMM fits across 500 deterministic replicates, four data-generating cells, and two candidate mixed-effects architectures. All ten block manifests pass integrity verification, and every M1/M2 pair uses the same generated dataset.

Primary result: M1 satisfies the prespecified null Type-I and 95% coverage compatibility screens under both tested random-effect correlation conditions. M2 does not. The locked architecture rule therefore promotes M1 as the candidate for a future prospective design.

This result is deliberately narrower than a claim of universal validity. Exact-binomial nonrejection means no statistically detected departure from nominal calibration at Monte Carlo resolution N=500 under the tested DGP. It is not an equivalence proof, and Phase 7 does not establish that a scientific Self effect exists in live model data.

1. Scope and research boundary

This document is a Design Lab validation artifact. It does not edit or supersede the frozen Self-Referential Processing Pre-Registration v8.6.3. The frozen Gate 5 confirmatory estimator remains the item-level linear probability model unless a new preregistration is filed. Phase 7 evaluates a candidate GLMM analysis architecture for a future prospective design.

2. Locked Phase 7 design

• Four DGP cells cross beta_Self = 0.00 or 0.10 log-odds with true rho_model = 0.00 or 0.30.

• M1 uses a correlated model intercept/Self-slope random-effects block; M2 uses the corresponding uncorrelated block. Both include an item random intercept.

• Each generated dataset contains 100 models x 400 items = 40,000 observations with exact within-model Self balance.

• 500 deterministic replicates x 4 DGP cells x 2 fit architectures = 4,000 GLMM fits.

• Primary operating characteristics use all successful/evaluable fits, including singular fits when required inferential quantities are returned.

• Type-I and coverage compatibility are tested with prespecified two-sided exact binomial tests. p >= 0.05 is required. Power is descriptive only.

3. Recovery and final integrity

A post-aggregation diagnostic on the initial run found that Blocks 02-10 contained result records but no GLMM estimates because the R runtime lacked the lme4 package. Those blocks were archived. The dependency was restored, and only the originally failed prespecified blocks were rerun with unchanged frozen Phase 7 sources and original deterministic seeds. Block 01 was preserved. A partial Block 01 aggregate had been accidentally observed before recovery; no Phase 7 statistical specification or source code was changed afterward.

Integrity check	Final value
raw_rows	4000
unique_reps	500
unique_cells	4
fit_specs	['M1', 'M2']
block_manifests	10
all_block_integrity_pass	True
M1_M2_dataset_sha_pairing	True
expected_total_fits	4000
integrity_pass	True

4. Primary operating characteristics

Cell	Spec	Reject	Cover	Mean est.	Bias	Singular	Status
A_NULL rho=0	M1	6.6%	93.4%	0.00027	0.00027	90.0%	PASS
A_NULL rho=0	M2	7.2%	92.8%	0.00035	0.00035	85.6%	FAIL
B_NULL rho=30	M1	6.4%	93.6%	0.00019	0.00019	87.0%	PASS
B_NULL rho=30	M2	7.4%	92.6%	0.00028	0.00028	83.8%	FAIL
C_EFFECT rho=0	M1	99.6%	94.2%	0.10040	0.00040	89.2%	Cov PASS
C_EFFECT rho=0	M2	99.8%	93.4%	0.10049	0.00049	83.6%	Cov PASS
D_EFFECT rho=30	M1	99.6%	94.2%	0.10042	0.00042	88.6%	Cov PASS
D_EFFECT rho=30	M2	99.6%	93.8%	0.10049	0.00049	82.8%	Cov PASS

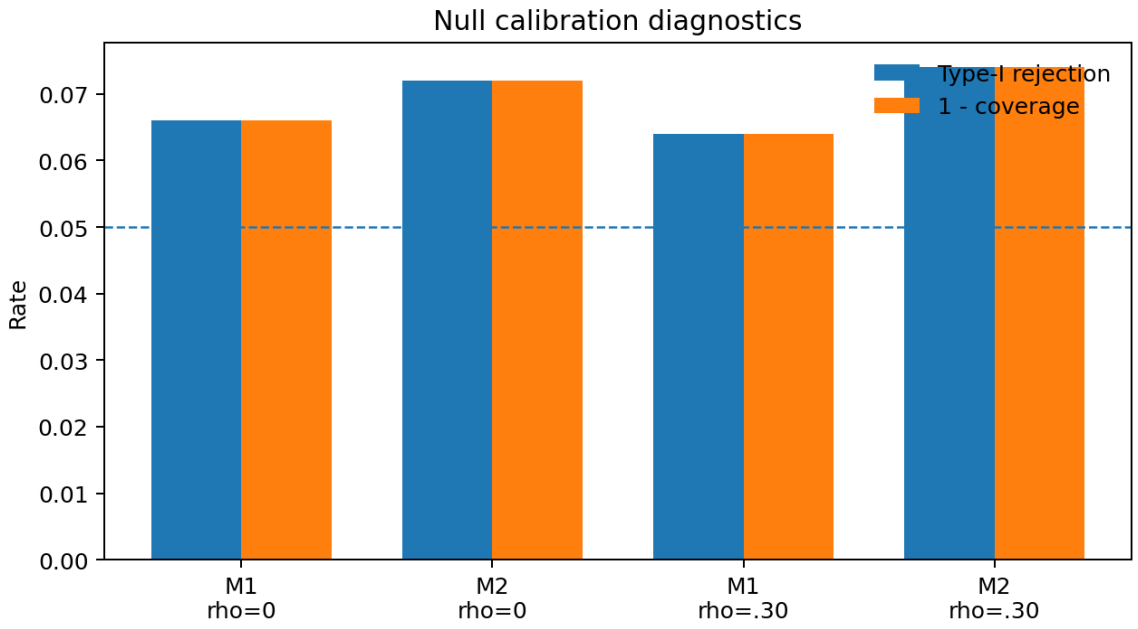
Null calibration detail

M1, rho=0.00: Type-I 6.6% (exact p=0.1007); coverage 93.4% (exact p=0.1007).

M1, rho=0.30: Type-I 6.4% (exact p=0.1504); coverage 93.6% (exact p=0.1504).

M2, rho=0.00: Type-I 7.2% (exact p=0.0305); coverage 92.8% (exact p=0.0305).

M2, rho=0.30: Type-I 7.4% (exact p=0.0179); coverage 92.6% (exact p=0.0179).



5. Locked architecture decision

The prespecified rule is architecture-independent: if only one architecture demonstrates acceptable null calibration and confidence-interval coverage across both tested covariance-generating conditions, that architecture becomes the candidate for the prospective design.

M1 passes both null Type-I screens and both corresponding null coverage screens. M2 fails both null Type-I screens and both corresponding null coverage screens. Therefore M1 is the Phase 7 candidate architecture and M2 is not promoted.

The correct interpretation is: no statistically detected departure from nominal calibration at resolution N=500 under the tested DGP. This is not an equivalence proof or a guarantee outside the simulated conditions.

6. Effect recovery and power

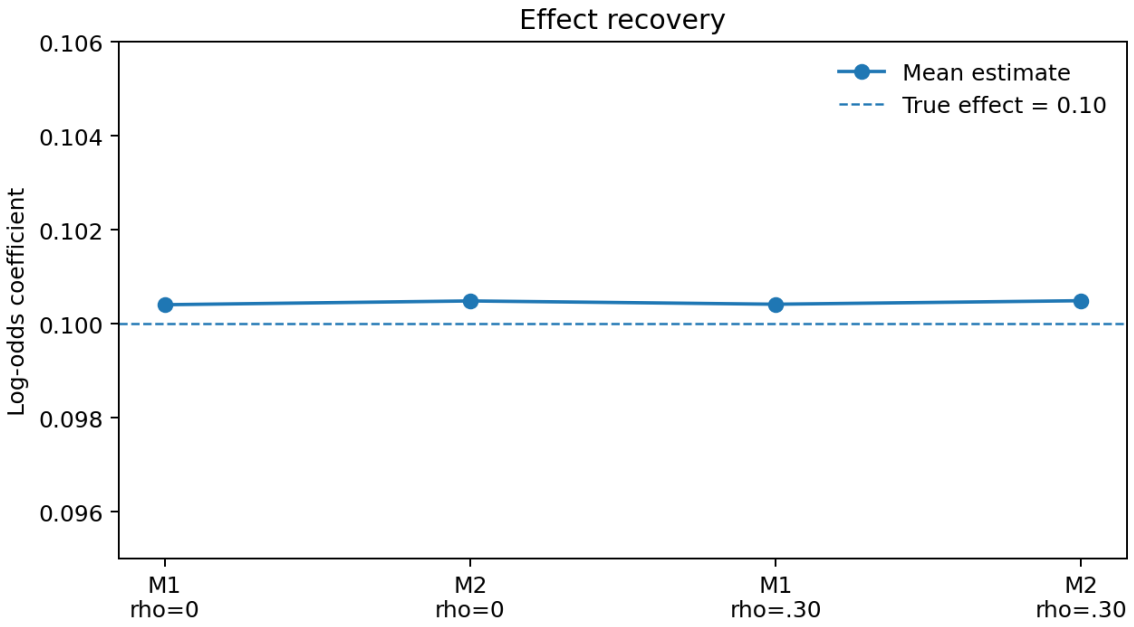
P7_C_EFFECT_RHO0, M1: mean estimate 0.100405; signed bias 0.000405; relative bias 0.405%; rejection 99.6%; coverage 94.2%.

P7_C_EFFECT_RHO0, M2: mean estimate 0.100485; signed bias 0.000485; relative bias 0.485%; rejection 99.8%; coverage 93.4%.

P7_D_EFFECT_RHO30, M1: mean estimate 0.100415; signed bias 0.000415; relative bias 0.415%; rejection 99.6%; coverage 94.2%.

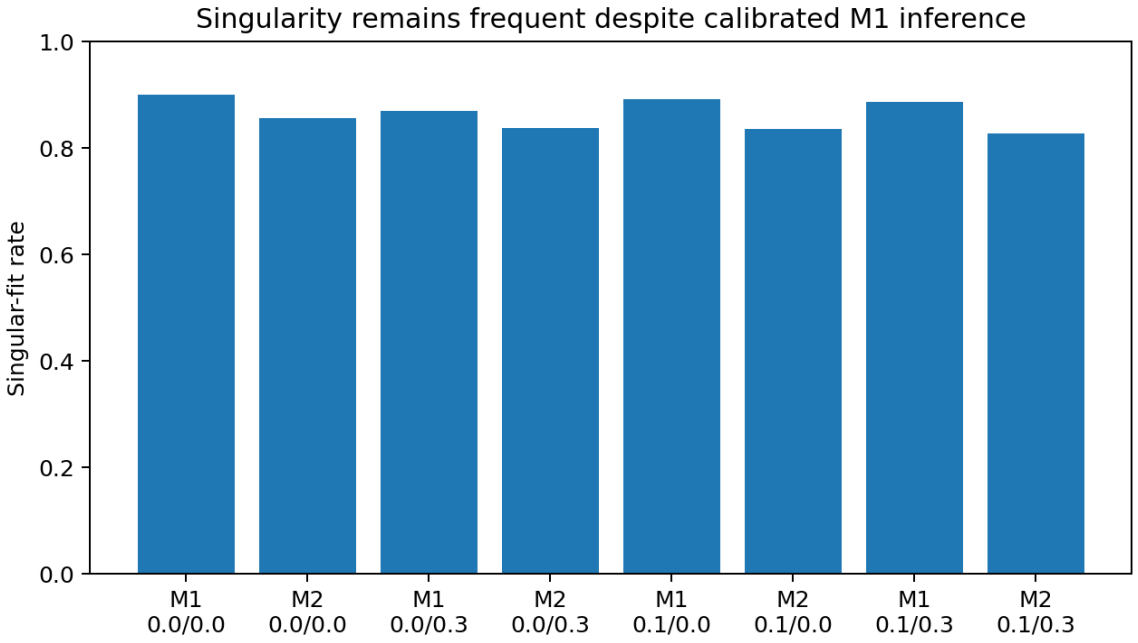
P7_D_EFFECT_RHO30, M2: mean estimate 0.100490; signed bias 0.000490; relative bias 0.490%; rejection 99.6%; coverage 93.8%.

Power is reported because it was prespecified, but power is not a validity/calibration acceptance criterion.



7. Singularity and paired diagnostics

Singularity remains common in both architectures. This confirms the earlier Design Lab diagnosis that the small random-effect structure is weakly identified even at high information scale. Phase 7 therefore asks a narrower question: whether the fixed Self coefficient retains acceptable operating characteristics when singular fits are retained as prespecified.



Cell	Mean M2-M1 est.	Mean M2-M1 SE	M1 singular	M2 singular
P7_A_NULL_RHO0	0.000086	-0.000346	90.0%	85.6%
P7_B_NULL_RHO30	0.000086	-0.000368	87.0%	83.8%
P7_C_EFFECT_RHO0	0.000081	-0.000379	89.2%	83.6%
P7_D_EFFECT_RHO30	0.000075	-0.000356	88.6%	82.8%

Paired fixed-effect differences are extremely small. M2 lowers singularity modestly but fails the locked null calibration screen; architecture selection therefore follows calibration rather than warning count or singularity frequency.

8. Limitations and interpretation

• Results are conditional on the locked Phase 7 DGP, information scale, effect sizes, variance components, assignment geometry, and software implementation.

• High singularity remains a structural warning about random-effect identifiability even though M1 fixed-effect operating characteristics passed the prespecified calibration screen.

• Exact-binomial nonrejection is not equivalence. It means no departure was detected at the chosen Monte Carlo resolution.

• Phase 7 validates an analysis architecture under simulation; it does not establish that a Self effect exists in live model data.

• M1 cannot be substituted retroactively into the frozen v8.6.3 Gate 5 confirmatory analysis. A future confirmatory design using M1 should be preregistered prospectively.

9. Reproducibility and frozen hashes

cce97faba969a6da873d9a5a34d9da6421fdf24d1a04df29c1a599fb3044e3d0 phase7_results.csv

e77755e76a3fa75ed8a130e578b864af75ca2b42a614f84306bc9b097f5e605e phase7_summary.csv

17c16f1279a72f2916a97544586e7a6095699281a3a21bcdfd3c0b21735ff1eb phase7_paired.csv

9ee3720597b35c553f0b3dd7a6f2ca159b904609b171cd41b4f0800ab2f8eae2 phase7_blocks_manifest.csv

ae92741d95ce01c46b7b197cc7d18386a3ab87494fdf2c1f506682b270b6c318 phase7_integrity_report.txt

7c33848597dd1fd928b600369ccf2ce11b483eeeedbf97f0e62b66ee8fcfce0d phase7_design_lock.md

1f9192b4e1f0d9bf21b90191165f2b83c7c0a8286856dd7e6d105a2f737717fe phase7_inferential_validation_runner.py

ac3f128af92a10fbdb1ca627a9f0e6af0ac032a6587155a47f6dc8280f187530 phase7_glmer_fit.R

5aec0011ee756d7b3657ff07f733cd58170d5b3f2076332ed5d05c404f7a7de2 phase7_aggregate.py

The release bundle also includes the ten block-level result/manifests and the original recovery incident record so the recovery history remains auditable rather than silently erased.

10. Final conclusion

Phase 7 is complete. The recovered 4,000-fit production run passes structural integrity. Under the frozen decision rule, M1 is the candidate architecture for a future prospective design; M2 is not promoted. The result supports moving to a new preregistration built around M1, while preserving the frozen v8.6.3 analysis unchanged.